Lecture 1 What is Abstract Algebra?

- **The Well-Ordering Principle**: any nonempty subset of the natural numbers has a least element.
- **Bézout's Lemma**: if $a, b, d \in \mathbb{Z}$ such that $\gcd(a, b) = d$ then there exists $x, y \in \mathbb{Z}$ such that $ax + by = d$. Moreover, d is the smallest positive integer that can be expressed in this form.
  Corollary. If $a, b, n \in \mathbb{Z} \setminus \{0\}$ such that $\gcd(a, b) = 1, a \mid n, b \mid n$ then $ab \mid n$.
- **Euclid's Lemma**: let $p \in \mathbb{Z}, p \geq 2$, then p is prime iff $\forall a, b \in \mathbb{Z}$ if $p \mid ab$ then $p \mid a$ or $p \mid b$.
- **Theorem 1.23**: let $a, b, m \in \mathbb{Z}, m > 0$. If $\gcd(a, m) = 1$ then $ax \equiv b \mod m$ has a solution. Moreover, this solution is unique up to congruence modulo m.
- **Fermat's Little Theorem**: if p is prime then $a^{p-1} \equiv 1 \mod p$ or $a^p \equiv a \mod p, \forall a \in \mathbb{Z}$.
- **The Chinese Remainder Theorem**: let $m_1, \ldots, m_r$ be pairwise relatively prime positive integers and let $a_1, \ldots, a_r$ be any integers. Then the system of linear congruences:

$$\begin{aligned} x &\equiv a_1 \mod m_1 \\ \vdots \quad & \quad \vdots \\ x &\equiv a_r \mod m_r \end{aligned}$$

  has a unique solution modulo $M = m_1 \ldots m_r$.

Lecture 2 Permutations

- Given a set X, the **symmetric group** on X, denoted S_X, is the set of all permutations of X.
  The **product** of two permutations is the permutation formed through composition.
  Permutations are not commutative.
- Let $x_1, x_2, \ldots, x_r \in [n]$ be distinct elements with $1 \leq r \leq n$, the r**-cycle** $(x_1\ x_2\ \ldots\ x_r)$ is the permutation from S_n that maps $x_1 \mapsto x_2, \ldots, x_r \mapsto x_1$ and fixes all other elements.
  The identity permutation is typically denoted by (1).
  2-cycles are called **transpositions**.
  Two cycles are called **disjoint** if no elements are moved by both cycles.
  If two permutations $\pi, \tau \in S_n$ are disjoint, then they commute.
- **Theorem 2.8**: Every permutation is either a cycle or a product of disjoint cycles. Moreover, this decomposition is unique up to order of the terms.
- $(x_1 x_2 \ldots x_r)^{-1} = (x_r x_{r-1} \ldots x_1)$
  If $\pi_1, \ldots, \pi_k \in S_n$, then $(\pi_1 \pi_2 \ldots \pi_k)^{-1} = \pi_k^{-1} \pi_{k-1}^{-1} \ldots \pi_1^{-1}$.
- Given two permutations $\pi, \tau \in S_n$, they have the same **cycle structure** if their decompositions have the same number of r-cycles $\forall r \leq n$.
  Theorem 2.15: two permutations $\pi, \tau \in S_n$ have the same cycle structure iff there exists some permutation $\alpha \in S_n$ such that $\tau = \alpha \pi \alpha^{-1}$. Then σ, τ are called **conjugates** and the operation of multiplying α, α^{-1} is known as **conjugation**.
- Given any permutation $\pi \in S_n$, the associated **permutation matrix** $P = \sum_{i=1}^{n} E_{\pi(i),i}$ where $E_{\pi(i),i}$ has a 1 in row $\pi(i)$, column i, and 0's everywhere else.
  $P^{-1} = P^T$
  If P is the permutation matrix of a transposition, then $\det(P) = -1$.
  $\det(P)$ is either 1 or -1.
- **Theorem 2.22**: for $\pi, \tau \in S_n$ with permutation matrices of P, T, the associated permutation matrix of $\pi\tau$ is PT.
- A permutation is said to be **even** if it can be expressed as a product of an even number of transpositions; **odd** if it can be expressed as a product of an odd number of transpositions.
  Theorem 2.24: a permutation $\pi \in S_n$ is even iff $\det(P) = 1$ and odd iff $\det(P) = -1$.
  Every permutation is even or odd but not both.

Lecture 3 Groups

- Let G be a nonempty set and $*$ be a binary operation on G. $(G, *)$ is a **group** iff
  1. (Closure): $\forall a, b \in G, a * b \in G$.
  2. (Associativity): $\forall a, b, c \in G, (a * b) * c = a * (b * c)$.
  3. (Identity): $\exists e \in G$ such that $\forall a \in G, a * e = e * a = a$. e is called the **identity element**.
  4. (Inverses): $\forall a \in G, \exists b \in G$ such that $a * b = b * a = e$. $b = a^{-1}$ is the **inverse** of a.
- (Cancellation Law): $\forall a, b, c \in G, ab = ac \Rightarrow b = c$ and $ba = ca \Rightarrow b = c$.
  (Unique Inverse): $\forall a, b, c \in G, ab = ba = e, ac = ca = e \Rightarrow b = c$.
  (Unique Identity): $\forall a \in G, ab = ba = a \Rightarrow b = e$.
  $(ab)^{-1} = b^{-1}a^{-1}$
  $\left(a^{-1}\right)^{-1} = a$
- Let G be a group, $a, b \in G$. Then the equations $ax = b, xa = b$ have unique solutions in G.
- If $\forall a, b \in G, ab = ba$ then G is **abelian**. Otherwise, G is **non-abelian**.
- The **powers** a^n for $n \geq 0$ is defined inductively by $a^0 = 3, a^{n+1}a = a^n a$. Furthermore, $a^{-n} = \left(a^{-1}\right)^n$ for $n > 0$.
  $\forall m, n \in \mathbb{Z}, a^{m+n} = a^m a^n, a^{mn} = (a^m)^n$
  If G is abelian then $\forall n \in \mathbb{Z}, (ab)^n = a^n b^n$
- If $|G| = n \in \mathbb{Z}^+$ then the group G has **order** n. Otherwise G has **infinite order**.
  If $a^k = e$ for some $k \in \mathbb{Z}^+$ then the smallest such k is the **order** of a ($|a| = k$). Otherwise a has **infinite order**.
- Let G be a group with $|G| = n \in \mathbb{Z}^+$, then $\forall a \in G, |a| \leq n$.
  Let G be a group and $a \in G$ with $|a| = n \in \mathbb{Z}^+$, then $\forall k \in \mathbb{Z}$, if $a^k = e$ then $n \mid k$.

Lecture 4 Groups

- Let G be a group and $H \subseteq G$ be a nonempty subset. H is a **subgroup** of G ($H \leq G$) iff
  1. (Identity): $e \in H$
  2. (Closure): $\forall x, y \in H, xy \in H$
  3. (Inverse): $x \in H \Rightarrow x^{-1} \in H$

  $\{e\}$ is the trivial subgroup.
  Theorem 3.20: $H \leq G \Rightarrow H$ is also a group.
- Let G be a group and $a \in G$. Define $\langle a \rangle = \{a^k | k \in \mathbb{Z}\}$. Then $\langle a \rangle \leq G$ and $|\langle a \rangle| = |a|$.
- **Proposition (The Finite Subgroup Test)**: Let G be a finite group with $|G| = n|$ and let $H \subseteq G$ be nonempty. Then $H \leq G$ iff $\forall a, b \in H, ab \in H$.
  Theorem (One-Step Subgroup Test): Let G be a group and $H \subseteq G$ be nonempty. Then $H \leq G$ iff $\forall a, b \in H, ab^{-1} \in H$.
- **Group presentations**: for a nonempty subset $X \subseteq G$, the **smallest subgroup containing** X **/ subgroup generated by** X is

  $$\langle X \rangle = \text{ the intersection of all subgroups containing } X = \bigcap_{X \subseteq H \leq G} H$$

  The elements of X are the **generators** of the group $\langle X \rangle$.
  Lemma 3.26: let $\mathcal{H}$ be a family of subgroups of G. Then $\bigcap \mathcal{H} = \bigcap_{H \in \mathcal{H}} H \leq G$, i.e. the intersection of an arbitrary number of subgroups is still a subgroup.
- **Lagrange's Theorem**: let G be a finite group with $|G| = n$. If $H \leq G$ with $|H| = k$ then $k \mid n$.
- Let G be a group and $a \in G$, then the subgroup $\langle a \rangle$ is the **cyclic subgroup generated by** a. If $G = \langle a \rangle$ then G is a **cyclic group**.
  If the binary operation is addition then $\langle a \rangle = \{na | n \in \mathbb{Z}\}$.
  A cyclic group can have more than one generator.

- **Theorem**: every cyclic group is abelian.
  Proposition: Every subgroup of a cyclic group is cyclic.

- **Fundamental Theorem of Cyclic Groups**: suppose G is a finite cyclic group with $G = \langle a \rangle$ and $|a| = n$, then for each $d \mid n$ there exists a unique subgroup $H \leq G$ such that $|H| = d$.

Lecture 5 Groups

- Given $X \subseteq G$ with $b \in X$, any equation of the form $b = e$ is a **relation on** X.
  Given a set of generators X of a group G, along with a set of relations R, $\langle X \mid R \rangle$ is a **presentation** of the group G.
  - finite case: for the cyclic group of order n, $C_n = \langle a | a^n = e \rangle$ is the group generated by a such that $|a| = n$, C_n is the largest group generated by a single element satisfying $a^n = e$
  - infinite case: for the group generated by a single element of infinite order, $C_\infty = \langle a \mid \rangle = \langle a \rangle$ is the largest group in which every element can be written as an integer power of a single element.

  $\langle a, b \rangle$ is the **free group** generated by 2 elements that contains all of the finite length words that can be expressed using a, b, a^{-1}, b^{-1}.

- **Theorem**: any group G can be presented in terms of generators and relations.
  A group is said to be **finitely presented** if it has a presentation with a finite number of generators and relations.

- The group $D_n = \langle x, y | y^2 = e, x^n = e, xy = yx^{-1} \rangle$ is the nth **Dihedral Group**.
  $|D_n| = 2n$.

- Given two groups G, G', a function $\varphi : G \to G'$ is a **homomorphism** iff $\forall a, b \in G, \varphi(ab) = \varphi(a)\varphi(b)$.

- **Proposition 3.54**: let G, H be groups and $\varphi : G \to H$ be a homomorphism. Then
  1. $\varphi(e_G) = e_H$
  2. $\forall g \in G, \varphi(g^{-1}) = \varphi(g)^{-1}$
  3. $\forall n \in \mathbb{Z}, \varphi(g^n) = \varphi(g)^n$

- **Proposition**: let $\varphi : G \to H$ be a homomorphism. If $g \in G$ has $|g| = n$ then $|\varphi(g)|$ divides n.

- Let $\varphi : G \to H$ be a homomorphism.
  1. If φ is injective then it's a **monomorphism**
  2. If φ is surjective then it's an **epimorphism**
  3. If φ is bijective then it's an **isomorphism** ($G \cong H$)
  4. If $H = G$ and φ is an isomorphism then φ is an **automorphism**

- **Proposition**: $\mathbb{Z} \cong C_\infty$ and $\mathbb{Z}/n\mathbb{Z} \cong C_n, \forall n \in \mathbb{Z}^+$.

Lecture 6 Groups

- **Theorem 3.58** A homomorphism $\varphi : G \to H$ is an isomorphism iff there exists an inverse homomorphism φ^{-1}.
  Corollary: if $\varphi : G \to H$ is an isomorphism then $|g| = |\varphi(g)|$.

- If $\varphi : G \to H$ is an isomorphism then the **kernel of** φ is $ker\varphi = \{x \in G | \varphi(x) = e\}$. The **image of** φ is $im\varphi = \{h \in H | \exists x \in G \text{ s.t. } h = f(x)\}$.

- **Proposition**: a homomorphism $\varphi : G \to H$ is a monomorphism iff $ker\varphi = \{e\}$.
  Theorem 3.64: let $\varphi : G \to H$ be a homomorphism, then $ker\varphi \leq G$ and $im\varphi \leq H$.

- **Theorem 3.66**: let G be a group and $g \in G$. Define $\gamma_g : G \to G$ by $\gamma_g(a) = gag^{-1}$ for all $a \in G$, then γ_g is an automorphism, i.e. for any group G, conjugacy is an automorphism.

- Let G be a group, $H \leq G, g \in G$. Then $gH = \{gh | h \in H\}$ is a **left** H**-coset** and $Hg = \{hg | h \in H\}$ is a **right** H**-coset**.

- Let $H \leq G$. The **index** of H in G, $[G : H]$, is the number of distinct (left) cosets of H in G, which is either a positive integer or ∞.
  Theorem: Let $H \leq G, a, b \in G$, then:
  1. $|aH| = |H|$
  2. $aH = bH \Leftrightarrow b^{-1}a \in H$
  3. $aH = H \Leftrightarrow a \in H$
  4. $aH = bH$ or $aH \cap bH = \varnothing$
  5. $\bigcup_{a \in G} aH = G$
- **Lagrange's Theorem**: if G is a group of finite order and $H \leq G$ then $|G| = [G : H]|H|$ and hence $|H| \mid |G|$.

Lecture 7 Groups

- A subgroup $N \leq G$ is **normal** in G ($N \lhd G$) iff $gN = Ng, \forall g \in G$.
- **Theorem**: let $N \leq G$, then the following are equivalent:
  1. $N \lhd G$
  2. $gNg^{-1} = N, \forall g \in G$
  3. $gNg^{-1} \subseteq, \forall g \in G$ (standard normality proof)
  4. N is the kernel of a homomorphism
  5. The set of N-cosets forms a group
- **Theorem 4.19**: if $\varphi : G \to G'$ is a homomorphism then $ker\varphi \lhd G$.
- **Theorem**: if $N \leq G$ and $[G : N] = 2$ then $N \lhd G$.
- The **center** of a group G is defined as $Z(G) = \{z \in G | \forall g \in G, gz = zg\} = \{z \in G | \forall g \in G, gzg^{-1} = z\}$ (set of elements fixed by all conjugations).
  $Z(G) \leq G$ and $Z(G) \lhd G$ sicne $gxg^{-1} = x, \forall x \in Z(G)$.
- A nontrivial group G is called **simple** iff its only normal subgroups are $\{e\}$ and G.
- Let $(G, *), (H, \cdot)$ be two groups. The **external direct product** of G and H has underlying set $G \times H$ with the group operation of component-wise multiplication.
  $G \times H$ is a group with identity element (e_G, e_H).
  $G \times H \cong H \times G$
- **Theorem 4.6** $G \times H$ is abelian iff both G, H are abelian.
- Let $\varphi : H \to Aut(G)$ be a homomorphism. The **semidirect product** of G and H under φ, $G \rtimes_\varphi H$, has underlying set $G \times H$ with the operation $(g_1, h_1)(g_2, h_2) = (g_1(\varphi(h_1)(g_1)), h_1h_2)$.
  $G \rtimes_\varphi H$ is a group with identity (e_G, e_H).
  If φ is the trivial homomorphism then $G \rtimes_\varphi H$ is the direct product.
- Let G be a group adn $H, K \leq G$. The **subset product** is defined as $HK = \{hk | h \in H, k \in K\}$.
- **Theorem 4.34**: let G be a group and $H, K \leq G$ be finite subgroups. then $|HK| = \frac{|H||K|}{|H \cap K|}$.
  HK is not necessarily a subgroup of G.
  Proposition: $HK \leq G$ iff $HK = KH$.

Lecture 8 Groups

- **Theorem 4.35**: $H \leq G, N \lhd G \Rightarrow HN \leq G$.
  Proposition: $H, N \lhd G \Rightarrow HN \lhd G$.
- Let G be a group and $H, N \leq G$.
  1. If $N \lhd G, H \cap N = \{e\}$ then HN is an **internal semidirect product**.

2. If $H, N \lhd G, H \cap N = \{e\}$ then $HN \lhd G$ is an **internal direct product** $(H \oplus N)$

- **Proposition**: let $N, K \lhd G$ with $N \cap K = \{e\}$, then $N \times K \cong N \oplus K$.
  Proposition: let $N, K \leq G$ with $N \lhd G$ and $N \cap K = \{e\}$, and let $\varphi : K \to Aut(N)$ be the homomorphism defined by $\varphi(k) = \gamma_k$. Then $N \rtimes_\varphi K \cong NK$.
- **Theorem**: let G be a group and $N \leq G$, then $N \lhd G$ iff the operation $(aN)(bN) = (ab)N$ is well-defined $\forall a, b \in G$ where $(aN)(bN) = \{a_1 b_1 | a_1 \in aN, b_1 \in bN\}$.
- **Theorem 4.39**: if $N \lhd G$ then the operation defined makes the set of cosets of N into a group.
- If $N \lhd G$ then the **quotient group** of G by N is the group of cosets under the operation $(aN)(bN) = abN$: $G/N =$ "G modulo N" $= \{aN | a \in G\}$.

Lecture 9 Groups

- **Proposition (Basic Properties of Quotient Group)**: Let $N \lhd G$,
  (1) $|G|$ finite $\Rightarrow |G/N| = |G|/|N|$
  (2) N is the kernel of the natural epimorphism $\eta : G \to G/N$ defined by $\eta(g) = gN$
- **Corollary**: $N \lhd G \Leftrightarrow N$ is the kernel of some homomorphism with domain G.
- **Theorem 4.48**: let $K \lhd G$. $X \leq G/K$ iff $X = N/K$ for wome $K \leq N \leq G$.
- **First Isomorphism Thoerem**: if $\varphi : G \to H$ is a homomorphism then $G/ker\varphi \cong im\varphi$.

Lecture 10 Groups

- **Second Isomorphism Theorem**: let $H, K \leq G$ such that $K \lhd G$. Then $K \lhd HK$ and $H/H \cap K \cong HK/K$.
- **Third Isomorphism Theorem**: if $H, K \lhd G$ with $K \leq H$ then $H/K \lhd G/K$ and $(G/K)/(H/K) \cong G/H$.
- **Correspondence Theorem**: given $K \lhd G$ there is a 1-to-1 correspondence between subgroups of G containing K and subgroups of G/K. Moreover, if $K \leq H, K \leq N$ then
  (1) $H \leq N \Leftrightarrow H/K \leq N/K$ and $[N : H] = [N/K : H/K]$
  (2) $H \lhd N \Leftrightarrow H/K \lhd N/K$ and $N/H \cong (N/K)/(H/K)$

Lecture 11 Groups Actions

- **Definition 1**: If X is a nonempty set and G is a group then G **acts on** X if there exists a function $\alpha : G \times X \to X$ called an **action** such that
  (1) $\alpha(e, x) = x, \forall x \in X$
  (2) $\alpha(g, \alpha(h, x)) = \alpha(gh, x), \forall g, h \in G, \forall x \in X$

  Definition 2: given a group G and a nonempty set X, a **group action** is any homomorphism $\beta : G \to S_X$.
  Proposition: these two definitions are equivalent.
- **Proposition**: let G act on a set X, then for all $g \in G, x, y \in X$, if $G \cdot x = y$ then $x = g^{-1} \cdot y$.
- Let G act on X and $x \in X$, then **orbit** of x is defined as $\mathcal{O}(x) = \{g \cdot x | g \in G\} \subseteq X$. The **stabilizer** of x is defined as $G_x = \{g \in G | g \cdot x = x\} \subseteq G$. Orbits are equivalence classes of elements of X, so the orbits of X partition X, i.e. if X is finite then $|X| = \sum |\mathcal{O}(x_i)$ where one x_i is chosen from each orbit.
- **Proposition 5.8**: suppose G acts on a set X. The relation $x \sim y$ iff $\exists g \in G$ such that $g \cdot x = y$ is an equivalent relation.
- **Lemma 5.9**: let G act on X. For each $x \in X, G_x \leq G$.
- **Orbit-Stabilizer Theorem**: if G acts on X and $x \in X$ then $|\mathcal{O}(x)| = [G : G_x]$.
  If G is finite then the number of elements in any orbit is a divisor of $|G|$.

- **conjugacy class** of x: $x^G = \mathcal{O}(x) = \{gxg^{-1} | g \in G\}$
  centralizer of x in G: $C_G(x) = G_x = \{g \in G | gxg^{-1} = x\}$
  conjugacy class of H: $H^G = \mathcal{O}(H) = \{gHg^{-1} | g \in G\}$
  normalizer of H in G: $N_G(H) = G_H = \{g \in G | gHg^{-1} = H\}$
  $N_G(H)$ is the largest subgroup in which H is normal. If $H \lhd G$ then $N_G(H) = G$.

- **Proposition 5.16**: let G be a group, $g \in G, H \leq G$, then

  (1) $|g^G| = [G : C_G(g)]$
  (2) $|H^G| = [G : N_G(H)]$
  (3) If $x_1^G, \ldots, x_n^G, y_1^G, \ldots, y_k^G$ are distinct conjugacy classes of G such that $|y_i^G| = 1$ and $|x_i^G| > 1, \forall i$ then

$$|G| = |Z(G)| + \sum_{i=1}^{n} [G : C_G(x_i)] \qquad \textbf{(Class Equation)}$$

Lecture 12 Group Actions

- **Burnside's Lemma**: let X be a finite set acted on by a finite group G. Further, for $g \in G$ define $X_g = \{x \in X | g \cdot x = x\}$ and let r be the number of distinct orbits in X under G. Then $r|G| = \sum_{g \in G} |X_g|$.

- **Proposition**: for prime p, if $|G| = p^n$ for $n \neq 0$ then $Z(G)$ is nontrivial.

- **Corollary**: every group or ofrder p^2 is abelian.

- **Cayley's Theorem**: if G is a group then G is isomorphic to a subgroup of S_G. In particular, if $|G| = n$ then G is isomorphic to a subgroup of S_n.
  Strong Cayley's Theorem: let $H \leq G$ with $[G : H] = n$, then there exists homomorphism $\varphi : G \to S_n$ with $ker\varphi \leq H$. Moreover, if H contains no nontrivial normal subgroups from G then G is isomorphic to a subgroup of S_n.

- **Lemma**: if G is a group of even order then G contains an element of order 2.

- **Corollary 5.27**: let G be finite group and p be the smallest prime divisor of $|G|$. If $H \leq G$ with $[G : H] = p$ then $H \lhd G$.

Lecture 13 p-groups

- A group G is called a **p-group** iff every element has order p^k for some $k \in \mathbb{Z}$.

- **Fixed Point Congruence Lemma**: Define $X_G = \{x \in X | \forall g \in G, g \cdot x = x\} = \{x \in X | G_X = G\}$. Let G be a finite group such thtat $|G| = p^n$. Suppose G acts on a finite set X. Then $|X| \equiv |X_G| \mod p$.
  Class equation: $|X| = |X_G| + \sum [G : G_{x_i}]$

- **Cauchy's Theorem**: if G is a finite group and $p \mid |G|$ then G has an element of order p and thus a subgroup of order p.

- **Corollary**: every finite p-group has order p^n.

- A subgroup $H \leq G$ is called a **p-subgroup** if H is a p-group.

- **Lemma 6.9**: let G be a finite group and $H \leq G$ be a p-subgroup. Then $[N_G(H) : H] \equiv [G : H] \mod p$.

- **Corollary**: if $p \mid [G : H]$ where H is a p-subgroup then $H \lneq N_G(H)$.

- If G is a finite group with $|G| = p^r m$ such that $p \nmid m$ then a subgroup of order p^r is called a **Sylow p-subgroup** of G.

- **First Sylow Theorem**: let G be a finite group with $|G| = p^r m, (p, m) = 1, r \geq 1$, then

  (1) There exists $H \leq G$ such that $|H| = p^r$ (existence of Sylow p-subgroups)
  (2) $\forall 1 \leq n \leq r, \exists K \leq G$ such that $|K| = p^n$ (p-subgroups exist of all possible orders)
  (3) If $n < r$ then each subgroup of order p^n is normal in a subgroup of order p^{n+1}

- **Second Sylow Theorem**: let G be a group with $|G| = p^r m, (p, m) = 1, r \geq 1$, then

  (1) All Sylow p-subgroups are conjugate to each other

(2) Let P be a Sylow p-subgroup of G. P is the unique Sylow p-subgroup iff $P \lhd G$

Lecture 14

- n_p: the number of Sylow p-subgroups.
  Third Sylow Theorem: if $|G| = p^r m$ with $r \geq 1, \gcd(p, m) = 1$ then $n_p \mid m$ and $n_p \equiv 1 \mod p$.
- **Sylow Theorem Summary**: let $|G| = p^r m, r \geq 1, \gcd(p, m) = 1, p$ prime, then
  (1) G has a Sylow p-subgroup.
  (2) All Sylow p-subgroups are conjugate to each other.
  (3) $n_p \mid m$ and $n_p \equiv 1 \mod p$.
- **Proposition**: let p, q be primes with $p < q, q \not\equiv 1 \mod p$, then every group of order pq is cyclic.
- **Proposition**: if p is an odd prime then every group of order $2p$ is either cyclic or isomorphic to D_p.
- **Lemma 6.22**: if G has a unique Sylow p-subgroup for each p diving $|G|$ then G is isomorphic to the direct product of its Sylow subgroups.

Lecture 15 Group Classifications

- **The Fundamental Theorem of Finite Abelian Groups**: every finite abelian group is isomorphic to a direct product of cyclic groups of prime power order, and this factorization is unique up to the arrangement of factors.
  That is, if G is a finite abelian group then

$$G \cong \mathbb{Z}/p_1^{n_1}\mathbb{Z} \times \cdots \times \mathbb{Z}/p_k^{n_k}\mathbb{Z}$$

  where the p_i's are not necessarily distinct and $\prod p_i^{n_i} = |G|$.

Lecture 16 Rings

- Let R be a nonempty set with 2 binary operations. The algebraic structure $(R, +, \cdot)$ is a **ring** iff it satisfies the following nine axioms:

$$\left.\begin{array}{l} 1.\ \forall a, b \in R, a + b \in R \\ 2.\ \forall a, b \in R, a + b = b + a \\ 3.\ \forall a, b, c \in R, (a + b) + c = a + (b + c) \\ 4.\ \exists 0 \in R \text{ such that } \forall a \in R, a + 0 = a \\ 5.\ \forall a \in R, \exists b \in R \text{ such that } a + b = 0 \end{array}\right\} (R, +) \text{ is abelian group}$$

$$\left.\begin{array}{l} 6.\ \forall a, b \in R, a \cdot b \in R \\ 7.\ \forall a, b, c \in R, (a \cdot b) \cdot c = a \cdot (b \cdot c) \end{array}\right\} (R, \cdot) \text{ is closed and associative}$$

$$\left.\begin{array}{l} 8.\ \forall a, b, c \in R, a \cdot (b + c) = a \cdot b + a \cdot c \\ 9.\ \forall a, b, c \in R, (a + b) \cdot c = a \cdot c + b \cdot c \end{array}\right\} \text{left and right distributivity}$$

  If $a \in R$ has a multiplicative inverse then a is called a **unit**.
- Special types of rings: let $(R, +, \cdot)$ be a ring,
  - A **commutative ring** is one such that $\forall a, b \in R, ab = ba$.
  - A **ring with unity** is a ring with a multiplicative identity (distinct from 0).
  - A ring with unity is said to be a **division ring** if $\forall a \neq 0$ in R, $\exists b \in R$ such that $ab = ba = 1$.
  - An **integral domain** is a commutative ring with unity that has no zero divisors: $\forall a, b \in R, ab = 0 \Rightarrow a = 0$ or $b = 0$.
  - A **field** is a commutative divison ring: $R \setminus \{0\}$ with multiplication is an abelian group.

$$
\begin{tikzcd}
 & \text{Rings} \arrow[dl, dash, "ab = ba"'] \arrow[dr, dash, "unity"] & \\
\text{Commutative Rings} \arrow[d, dash, "\text{unity, } ab = 0 \Rightarrow a/b = 0"'] \arrow[drr, dash, "ab = ba, ab = 0 \Rightarrow a/b = 0"] & & \text{Rings with Unity} \arrow[d, dash, "a^{-1} \in R"] \\
\text{Integral Domains} \arrow[dr, dash, "a^{-1} \in R"'] & & \text{Division Rings} \arrow[dl, dash, "ab = ba"] \\
 & \text{Fields} &
\end{tikzcd}
$$

- **Theorem (Elementary Properties)**: let R be any ring, $a, b]inR, n \in \mathbb{Z}$, then
  (1) $a \cdot 0 = 0 \cdot a = 0$
  (2) $a \cdot (-b) = (-a) \cdot b = -(ab) = -1 \cdot ab$ if R has unity
  (3) $(-a)(-b) = ab$
  (4) $(na)b = a(nb) = n(ab)$
- **Lemma**: in any ring $R, \forall a, b \in R$, $(a+b)^2 = a^2 + b^2 + ab + ba$.
- **Binomial Theorem**: given a commutative ring R such that $a, b \in R, n \in \mathbb{Z}^{\geq 0}$ then

$$(a+b)^n = \sum_{i=1}^{n} \binom{n}{i} a^i b^{n-i}$$

- The **characteristic** of a ring R ($Char(R)$) is the smallest positive integer such that $\forall a \in R, na = \underbrace{a + a + \cdots + a}_{n\text{-times}} = 0$. If no such n exists, $Char(R) = 0$.
- Let R, R' be rings and $\varphi : R \to R'$ be a function. Then φ is a **ring homomorphism** iff $\forall a, b \in R$,
  - $\varphi(a+b) = \varphi(a) + \varphi(b)$
  - $\varphi(ab) = \varphi(a)\varphi(b)$

  Theorem (Basic Properties): assume $\varphi : R \to R'$ is a ring homomorphism, then
  (1) $\varphi(0) = 0$
  (2) $\varphi(-a) = -\varphi(a)$
  (3) $\varphi(na) = n\varphi(a), \forall n \in \mathbb{Z}$

  $ker\varphi = \{a \in R | \varphi(a) = 0\}$
  Theorem (More Basic Properties): assume $\varphi : R \to R'$ is a ring homomorphism,
  (1) If φ is nontrivial and R has unity then $\varphi(1) = 1$.
  (2) If φ is nontrivial and $a \in R$ is a unit then $\varphi(a^{-1}) = \varphi(a)^{-1}$.
  (3) $\varphi(a^n) = \varphi(a)^n, \forall n \in \mathbb{Z}$.

Lecture 17 Rings

- If $(R, +, \cdot)$ is a ring and $S \subseteq R, S \neq \varnothing$ then S is a **subring** of R ($S \leq R$) iff $(S, +, \cdot)$ forms a ring.
- **Theorem (Subring Test)**: if $S \subseteq R, S \neq \varnothing$ then $S \leq R$ iff $\forall a, b \in S$,
  - $a - b \in S$
  - $ab \in S$
- Let R be a ring and $I \subseteq R, I \neq \varnothing$. I is called an **(2-sided) ideal** iff

- I is an additive subgroup of R
- If $r \in r$ and $a \in I$ then $r \in I$ and $ar \in I$

- **Lemma (Kernels are Ideals)**: if $\varphi : R \to R'$ is a ring homomorphism then $ker\varphi$ is an ideal of R.
- **Lemma (Intersection of Ideals is an Ideal)**: if $\mathcal{F}$ is a family of ideals from R then $\bigcap \mathcal{F}$ is an ideal of R.
- For $X \subseteq R$, the **ideal generated by** X is the intersection of all ideals from R containing X. Equivalently, (X) is the smallest ideal of R containing X.
- **Theorem 7.29**: let R be a commutative ring and $X \subseteq R, a \in R$, then
  (1) $(a) = \{ra + na | n \in \mathbb{Z}, r \in R\}$
  (2) If R has unity then $(a) = \{ra | r \in R\}$.
  (3) If R has unity then $(X) = \{r_1x_1 + \cdots + r_nx_n | n \in \mathbb{Z}^+, r_i \in R, x_i \in X\}$.

Lecture 18

- **Theorem 7.30**: let R be a ring and $I \leq R$ be an ideal, the additive quotient group R/I is a ring under the operations

$$(r + I) + (s + I) = (r + s) + I$$
$$(r + I) \cdot (s + I) = (r \cdot s) + I$$

  Conversely, if $I \leq R$ is any subring such that the operations above are well-defined then I is an ideal of R. **Corollary**: let R be a ring. A subset $I \subseteq R$ is an ideal iff there exists a ring R' and a ring homomorphism $\varphi : R \to R'$ such that $I = ker\varphi$
- **1st Isomorphism Theorem**: let $\varphi : R \to R'$ be a ring epimorphism, then $R/ker\varphi \cong R'$.
- **2nd Isomorphism Theorem**: let $A \leq R$ be a subring and $I \leq R$ be an ideal. Then
  (1) Let $A + I = \{a + i | a \in A, i \in I\}$, then $A + I \leq R$.
  (2) I is an ideal of $A + I$.
  (3) $(A + I)/I \cong A/(A \cap I)$
- **3rd Isomorphism Theorem**: let R be a ring. If K and I are ideals of R with $K \leq I \leq R$ then $R/I \cong (R/K)/(I/K)$.
- **Correspondence Theorem**: let R be a ring and $I \leq R$ be an ideal, then there is a bijective correpsondence between the subrings of R/I and the subrings of R containing I.
- A proper ideal $I \lneq R$ is a **maximal ideal** of R iff for all ideals J of R, if $I \leq J \leq R$ then $I = J$ or $J = R$.
- **Proposition**: if $I \leq R$ is an ideal and $u \in I$ is a unit then $I = R$.
- **Lemma 8.4**: let R be a commutative ring with unity, then R is a field iff its only ideals are (0) and R itself.

Lecture 19, 20

- **Theorem 8.6**: let R be a commutative ring with unity and $M \in R$ be an ideal, then R/M is a field iff M is maximal.
- Let R be a commutative ring and $P \lneq R$ be a proper ideal. P is called a **prime ideal** iff $\forall a, b \in R$, if $ab \in P$ then $a \in P$ or $b \in P$.
- **Theorem 8.17**: let R be a commutative ring with unity and $P \leq R$ be an ideal. R/P is an integral domain iff P is a prime ideal.
- **Theorem 8.15**: if R is a commutative ring with unity then every maximal ideal is a prime ideal.
- Let R be a commutative ring and $a, b \in R$,
  - a **divides** b ($a \mid b$) iff $\exists c \in R$ such that $b = ac$.
  - a, b are **associates** iff $a \mid b$ and $b \mid a$ iff $\exists$ a unit $u \in R$ such that $a = bu$.

- An element $p \in R$ is called **prime** iff p is a nonzero nonunit such that $\forall a, b \in R, p \mid ab \Rightarrow p \mid a$ or $p \mid b$.
- An element $p \in R$ is called **irreducible** iff p is a nonzero nonunit such that $\forall a, b \in R, p = ab \Rightarrow a$ or b is a unit iff its only divisors are units and associates.

- **Proposition**: let R be an integral domain and $p \in R$ be nonzero. Then p is prime iff (p) is prime.
- If R is an integral domain and every ideal of R is principal then R is called a **Principal Ideal Domain (PID)**.
- **Theorem 8.24**:
  (1) If R is an integral domain, $p \in R$ is prime $\Rightarrow p$ is irreducible.
  (2) If R is a PID and $(p) \leq R$ is a nonzero ideal then (p) is maximal iff p is irreducible.
  (3) If R is a PID and $p \in R$ is irreducible $\Rightarrow p$ is prime.
  (4) If R is a PID and $(p) \leq R$ is a nonzero ideal then (p) is prime iff (p) is maximal.

  If R is a PID and $p \in R$ nonzero then TFAE: p is prime, p is irreducible, (p) is prime, (p) is maximal.
- An integral domain R is called a **unique factorization domain (UFD)** iff
  - (Existence): $\forall a \in R$ nonzero nonunit, a can be expressed as a product of irreducible elements in R.
  - (Uniqueness): $\forall a \in R$, if $a = \prod_{i=1}^{m} p_i = \prod_{j=1}^{n} q_j$ where the p_i, q_j are irreducible then $m = n$ and there exists a permutation $\pi \in S_n$ such that $p_i = q_{\pi(i)}$ are associates for $1 \leq i \leq n$, i.e. the factorization is unique up to associate irreducible factors and the order of the factors.
- **Proposition**: if R is a UFD and $p \in R$ is irreducible $\Rightarrow p$ is prime.
- A ring R satisfies the **Noether condition / ascending chain condition** on principal ideals iff for any increasing chain of principal ideals $(a_1) \leq (a_2) \leq \cdots \leq (a_n) \leq \ldots$, $\exists n \in \mathbb{Z}^+$ such that $(a_i) = (a_n), \forall i \geq n$.
- **Lemma 8.28**: if a ring R is a PID then R satisfies the Noether condition on principal ideals.
- **Theorem 8.30**: A ring R is a PID then it's a UFD.

Lecture 21

- An integral domain R is called a **Euclidean Domain (ED)** iff there exists a valuation function $\chi : R \setminus \{0\} \to \mathbb{Z}^+$ s.t.
  (1) $\forall a, b \in R \setminus \{0\}, \chi(a) \leq \chi(ab)$.
  (2) $\forall a \in R, b \in R \setminus \{0\}, \exists q, r \in R$ such that $a = bq + r$ and either $r = 0$ or $\chi(r) < \chi(b)$.
- **Proposition**: if R is a Euclidean Domain and $a \in R$ then a is a unit iff $\chi(a) = \chi(1)$.
- **Theorem**: every Euclidean Domain is a Principal Ideal Domain.
  Fields $\subsetneq$ EDs $\subsetneq$ PIDs $\subsetneq$ UFDs $\subsetneq$ IDs $\subsetneq$ Commutative rings with unity
- Let R be a commutative ring and $a, b, d \in R$. d is a **greatest common divisor** of a, b iff
  (1) $d \mid a \wedge d \mid b$
  (2) $\forall c \in R, c \mid a \wedge c \mid b \Rightarrow c \mid d$

  a, b are called **relatively prime** iff 1 is a GCD of a, b.
- **Theorem** (Properties of GCDs):
  (1) If R is a UFD and $a, b \in R$ then $\gcd(a, b)$ exists and is unique up to a unit multiple.
  (2) If R is a PID, $a, b \in R, d$ is a GCD of a, b then there exists $m, n \in R$ such that $am + bn = d$ (Bézout's Lemma).
  (3) If R is a UFD then (2) doesn't necessarily hold.
  (4) If R is a UFD, $a, b \in R$ are relatively prime and $c \in R$ such that $a \mid bc$ then $a \mid c$ (Euclid's Lemma).
- Let R be a ring and let x be an element not in R, then x is an **indeterminate** or **transcendental** over R if $\forall a_i \in R, n \in \mathbb{Z}^{\geq 0}, a_0 + a_1 x + \cdots + a_n x^n = 0 \Leftrightarrow a_i = 0, \forall i$.

- Let R be a ring and x be indeterminate over R. Define the set $R[x] = \left\{\sum_{i=0}^{n} a_i x^i | a_i \in R, n \in \mathbb{Z}^{\geq 0}\right\}$ then $R[x]$ forms a **polynomial ring in 1 variable** over R under the operations of polynomial addition and multiplication

$$\left(\sum_{i=0}^{m} a_i x^i\right) + \left(\sum_{i=0}^{m} b_i x_i\right) = \sum_{i=0}^{m} (a_i + b_i) x^i$$
$$\left(\sum_{i=0}^{m} a_i x^i\right) \cdot \left(\sum_{j=0}^{n} b_j x^j\right) = \sum_{k=0}^{m+n} \sum_{i+j=k} a_i b_j x^k$$

  Let R be a ring and $f \in R[x]$ with $f = \sum_{i=0}^{n} a^i x^i$ with $a_n \neq 0$,

  (1) the **degree** of f ($\deg(f)$) is defined as the highest power of x with a nonzero coefficient. $\deg(0) = -\infty$.
  (2) f is called **monic** iff $a_n = 1$.

- **Theorem**: If R is a commutative ring with unity / an integral domain / a UFD, then so is $R[x]$.

- **Theorem 9.13**: let F be a field, then $F[x]$ is a Euclidean Domain with $\chi(f) = \deg(f)$. Moreover, the quotient and remainder are unique in the division algorithm.
  Corollary: if F is a field then $F[x]$ is both a PID and a UFD.

Lecture 22

- **Corollary**: let F be a field and $I \leq F[x]$ be a nonzero ideal. For $g \in F[x], I = (g)$ iff g is a nonzero polynomial of minimal degree in I.

- **Proposition**: let R be a UFD and $a, b \in R$ nonzero. Then $\forall d, q \in R, d$ is a gcd of a, b iff d is a gcd of $a, b - qa$, i.e. $\gcd(a, b) = \gcd(a, b - qa)$ as sets.

- For $a \in R, f \in R[x], a$ is a **root** of f iff $f(a) = 0$.

- **Theorem 9.18**: if R is a commutative ring with unity and $f \in R[x]$ then a is a root of f iff $(x - a) \mid f$.

- **Corollary**:

  (1) Let R be a commutative ring with unity and $f \in R[x]$ with $\deg(f) > 1$. If f is irreducible then f has no roots.
  (2) Let R be an integral domain and $f \in R$ be nonzero with $\deg(f) = n$, then f has at most n-many roots.

- **Theorem**; let F be a field and $f, g \in F[x]$, then $\gcd(f, g)$ exists, is unique, and can be written as linear combination of f, g over $F[x]$.

- **Corollary**: let F be a field, $f, g \in F[x], d = \gcd(f, g)$, then $\forall p \in F[x], \exists a, b \in F[x]$ such that $p = af + bg$ iff $d \mid p$.

- **Corollary 9.25**: let F, F' be field with $F \leq F'$. Let $f, g \in F[x]$. If $d_F = \gcd(f, g)$ in $F[x]$ and $d_{F'} = \gcd(f, g)$ in $F'[x]$ then $d_F = d_{F'}$.

Lecture 23

- Let R be a UFD and $f = \sum_{i=0}^{n} f_i x^i \in R[x]$. The **content** of f ($c(f)$) is the set of GCDs of $(f_0, \ldots, f_n)$. f is called a **primitive polynomial** iff $1 \in c(f)$.

- **Proposition**: let R be a UFD, $a \in R, f \in R[x]$, then $c(af) = a \cdot c(f)$.

- **Gauss's Lemma**: let R be a UFD and $f, g \in R[x]$. If $c(f) = c(g) = 1$ then $c(fg) = 1$, i.e. the product of two primitive polynomials is primitive. More generally, in any UFD $c(f)c(g) = c(fg)$.

- Let R be an integral domain and $S = R \times R^*$. Define an equivalent relation $\sim$ on S by $(a, b) \sim (c, d)$ iff $ad = bc$. For $a \in R, b \in R^*$, let $\frac{a}{b} = [(a, b)]_\sim$. Define the **field of fractions** as $F(R) = S/\sim = \left\{\frac{a}{b} | (a, b) \in S\right\}$. Define the binary operations on $F(R)$

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$
$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}$$

- **Proposition**: if E is a field and $R \leq E$ then there exists a monomorphism $\varphi : F(R) \hookrightarrow E$.
- **Lemma 9.36**: let R be a UFD and $F(R)$ be its field of fractions. Let $f, g \in R[x]$ with f primitive. If $f \mid g$ in $F(R)[x]$ then $f \mid g$ in $R[x]$.
  Subclaim: there exists $k \in F(R)$ such that $kh \in R[x]$ and kh is primitive.
- **Rational Root Test**: let R be a UFD and $f \in R[x]$ with $f = \sum_{i=1}^{m} f_i x^i$ with $f_m \neq 0$. If $a, b \in R$ such that $\gcd(a, b) = 1$ and $f\left(\frac{a}{b}\right) = 0$ then $a \mid f_0$ and $b \mid f_m$.
- **Theorem 9.39**: let R be a UFD and $f \in R[x]$ be primitive. Then f is irreducible in $R[x]$ iff either f is an irreducible in $F(R)[x]$ or f is an irreducible constant from R.

Lecture 24

- **Theorem 9.40**: if R is a UFD then $R[x]$ is a UFD.
- A polynomial $f \in R[x]$ is called **degreewise irreducible** iff $\forall g, h \in R[x]$ if $f = gh$ then $\deg(g) = 0$ or $\deg(h) = 0$.
- **Mod P Irreducibility Test**: let R be a PID and $f \in R[x]$. If $\eta_p(f)$ is irreducible in $(R/(p))[x]$ and $\deg(f) = \deg(\eta_p(f))$ then f is degreewise irreducible in $R[x]$.
- **Eisenstein-Schöneman Criterion**: let R be an integral domain, $p \in R$ be prime, $f \in R[x]$ with $f = \sum_{i=1}^{n} a_i x^i, a_n \neq 0$. If $p \nmid a_n, p \mid a_{n-1}, \ldots, p \mid a_1, p \mid a_0, p^2 \nmid a_0$ then f is degreewise irreducible over R.
  Corollary: if $f \in \mathbb{Z}[x]$ and $p \in \mathbb{Z}^+$ is prime such that $p \nmid a_n, \mid a_i, i = 0, \ldots, n-1, p^2 \nmid a_0$ then f is irreducible over $\mathbb{Q}$.

Lecture 25

- A field E is called an **extension field** of a field R iff $F \leq E$.
- **Fundamental Theorem of Field Theory**: let F be a field and $f \in F[x]$ be nonconstant. Then there exists extension field E in which $f(x)$ has a root.
- A **vector space** over a field F is an abelian group V whose group operation is denoted as addition and there exists a function $* : F \times V \to V$ such that
  (1) $ab * x = a * (b * x), \forall a, b \in F, x \in V$
  (2) $(a + b) * x = a * x + b * x, \forall a, b \in F, x \in V$
  (3) $a * (x + y) = a * x + a * y, \forall a \in F, x, y \in V$
  (4) $1 * x = x, \forall x \in V$
- **Proposition**: if E is an extension field of F then E is a vector space over F.
- If E/F is an extension field, we define the **degree** of the extension ($[E : F]$) to be the dimension of E as a vector space over F.
- **Theorem 10.3**: let E/F and K/E be finite field extensions. Then K/F is a finite field extension and $[K : F] = [K : E][E : F]$.
- Let E/F be a field extension and $u \in E$,
  - $F(u)$ denotes the smallest subfield of E containing $F \cup \{u\}$.
  - $F[u]$ denotes the smallest subring of E containing $F \cup \{u\}$.
  - The **evaluation homomorphism** is the ring homomorphism $\varphi_u : F[x] \to E$ such that $\varphi_u(f) = f(u)$. Then $im\varphi_u \leq F(u)$ since $F(u)$ is closed under addition and multiplication.
  - u is called **algebraic over** F iff there exists a nonzero $f \in F[x]$ such that $f(u) = 0$, i.e. $ker\varphi_u \neq \{0\}$. Otherwise, u is called **transcendental** over F.
  - E is called an **algebraic extension** of F iff every element of E is algebraic over F. Otherwise E is a **transcendental extension**.
- **Theorem**: let E/F be a field extension with $u \in E$,

(1) if u is transcendental over F then $F(u) \cong F(x) = Frac(F[x])$.

(2) if u is algebraic over F then $F(u) \cong F[x]/(p)$ for some irreducible $p \in F[x]$. Moreover, $n = [F(u) : F] = \deg(p)$ and $\{1, u, \ldots, u^{n-1}\}$ is a basis for $F(u)/F$.

Corollary: if u is algebraic over F then $F(u) = F[u]$.

- If E/F is a field extension and $u \in E$ is algebraic over F then we define the **minimal polynomial** of u to be the unique monic irreducible polynomial $p \in F[x]$ such that $p(u) = 0$.